# ECON 0100 | Vignette D2 | Commons

*Due in Recitation. Vignettes are a certificate of the work done together in Recitation.*

## Late Night Ingredients

Two skilled potion makers, Slughorn and Snape, can each brew different types of potions. They share access to a garden where they gather rare ingredients for their potions. The ingredients can only be harvested late at night, making it difficult to know who else uses the garden. Each must decide how much of the ingredients to harvest. The payoff matrix shows their monthly profits in Coins:

| | | **Snape** | |
|---|---|---|---|
| | | S | OH |
| **Slughorn** | Sustainable (S) | 8, 8 | 2, 12 |
| | Over-Harvest (OH) | 12, 2 | 4, 4 |

a) What is Slughorn's best response when Snape chooses Sustainable? _____

b) What is Slughorn's best response when Snape chooses Over-harvest? _____

c) What is *(are)* the Nash Equilibrium? _____

d) What is the socially efficient outcome? _____

# ECON 0100 | Vignette D3 | Public Goods

## The Broom Shed Roof

Oliver Wood and Cedric Diggory, the Gryffindor and Hufflepuff Quidditch captains, each have 10 coins in their team fund. The broom shed the two teams share needs a new roof. Each captain chooses to contribute all 10 coins (C) or nothing (DC). Madam Hooch collects the money by owl over the holidays: each captain sends a sealed pledge, and she opens both on the first day of term, so neither knows what the other gave until it's too late to change. Every 10 coins contributed buys repairs worth 6 coins to each team, whether or not that team contributed. Each team's payoff is the coins it keeps plus the value of the repairs. Fill in the payoff matrix below.

| | | **Diggory** | |
|---|---|---|---|
| | | C | DC |
| **Wood** | C | | |
| | DC | | |

a) Identify the best responses of Wood and Diggory.

b) Identify a Nash Equilibrium in this game if one exists.

c) Find the socially efficient strategies.

The pledges fall short, so Madam Hooch puts the roof to a vote of the two captains and the Head of School, Professor Dumbledore, whose budget also pays for the shed. A full new roof costs 21 coins, paid by a levy of 7 coins each on Gryffindor, Hufflepuff, and the school if a majority votes yes. Each voter's value of a new roof is below.

| **Wood** | **Diggory** | **Dumbledore** |
|---|---|---|
| 9 | 8 | 1 |

d) Would the levy receive a majority vote?

e) What is the marginal social benefit of the new roof?

f) Would building the roof be efficient?